Spatial weights from trade costs

1. **Gravity-disciplined trade costs**

$$\hat{\tau}_{ij} \propto \exp(\hat{\theta} T_{ij})$$

Estimated travel times discipline bilateral iceberg trade frictions.

2. **Baseline matrix: trade-cost exposure**

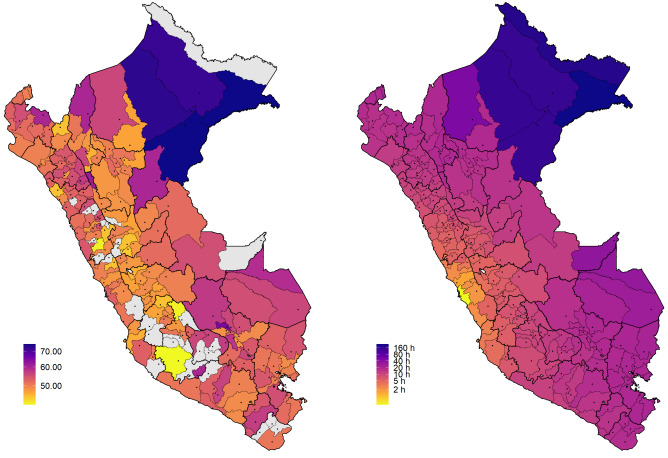
$$\omega_{ji}^{emp} = \frac{\hat{\tau}_{ji}^{1-\sigma}}{\sum_{k \neq j} \hat{\tau}_{jk}^{1-\sigma}}, \qquad \omega_{jj}^{emp} = 0.$$

Lower-cost links receive larger weights in the spatial lag.

3. **Market-size adjustments**
   - Y_i^0-adjusted matrix: larger initial markets receive more weight.
   - $(Y_i^0)^{1/2}$-adjusted matrix: keeps the market-size channel while reducing the dominance of Lima/Callao.

4. **Sensitivity to trade elasticity**
   - Baseline uses $\sigma = 4$.
   - Alternative matrices use $\sigma \in \{2, 3, 5, 8\}$.

LGP Prices and Travel Distance to Callao



(a) Median unit price of LPG (b) Travel Time from Callao

Notes: Median unit prices correspond to 2025 prices of 10kg cylinder containers. Calculations made using data from Osinergmin and OpenStreetMap.